

Reviewing “(A Very Brief) Introduction to Machine Learning”

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- We'll review some of the main introductory concepts related with Machine Learning using quizzes!
- Check 3.1.IntroToMachineLearning.pdf in PRADO.

Question 1

- What is the central idea of machine learning?
 - A. Writing explicit rules to describe the data
 - B. Using data plus learning algorithms to uncover underlying patterns
 - C. Building symbolic reasoning systems
 - D. None of the above

Question 2

- When can't we apply machine learning algorithms?
 - A. There is no pattern to learn, only randomness
 - B. We can describe our problem mathematically
 - C. We have no data whatsoever
 - D. We have data, but we know it's poorly annotated

Question 3

- Which of the following options most accurately represents the correct order from the highest to the lowest level of human supervision in the learning process?
 - A. Supervised learning → reinforcement learning → unsupervised learning
 - B. Reinforcement learning → supervised learning → unsupervised learning
 - C. Supervised learning → unsupervised learning → reinforcement learning
 - D. Unsupervised learning → reinforcement learning → supervised learning

Question 4

- Observe the video and tell the machine learning paradigm that you think it's more likely being used?

- A. Supervised Learning
- B. Reinforcement Learning
- C. Unsupervised Learning
- D. Self-supervised Learning



Question 5

- What do models “learn”/update in most cases??
 - A. Features
 - B. Labels
 - C. Weights/parameters
 - D. Only hyperparameters

Question 6

- Which model has no proper “training” phase in the traditional sense and mainly memorizes data?
 - A. Logistic regression
 - B. k-Nearest Neighbors (KNN)
 - C. Support Vector Machine
 - D. Decision Tree

Question 7

- Discriminative vs. Generative: which pairing is correct (assuming inputs X and outputs Y)?
 - A. Discriminative $\rightarrow p(X)$; Generative $\rightarrow p(Y|X)$
 - B. Discriminative $\rightarrow p(Y|X)$; Generative $\rightarrow p(X)$ or $p(X,Y)$
 - C. Discriminative $\rightarrow p(X,Y)$; Generative $\rightarrow p(Y|X)$
 - D. Both model $p(X)$ only

Question 8

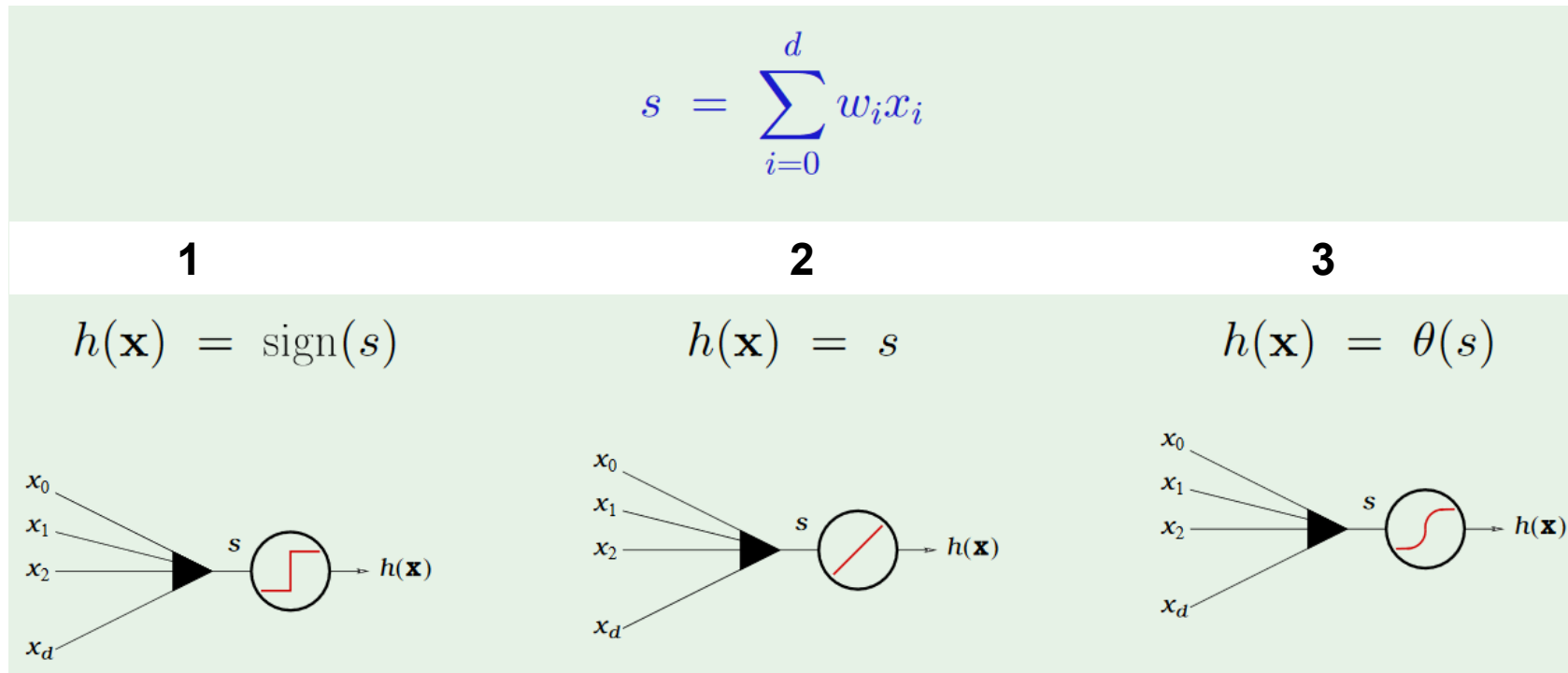
- In reinforcement learning, the learner mainly uses?
 - A. Exact labels for each state/situation
 - B. A reward or penalty signal to learn a policy
 - C. No label or feedback at all
 - D. None of the above

Question 9

- Machine learning training via gradient descent is framed as?
 - A. A combinatorial enumeration problem
 - B. An optimization problem minimizing a loss
 - C. A constraint satisfaction problem
 - D. A symbolic regression problem

Question 10

- Which model matches each graphical representation?
- A. Linear classification (Perceptron)
B. Linear regression
C. Logistic regression



Question 11

- Fill the table with the most common configuration of loss function and output units activation function.

Problem	Loss Function	Output units activation function
Regression	(a)	(d)
Binary Classification	(b)	(e)
Multiclass Classification	(c)	(f)

- 1 – Sigmoid
- 2 – Mean Squared Error
- 3 – Binary Crossentropy
- 4 – Softmax
- 5 – Identity/Linear
- 6 – Categorical Crossentropy

Question 12

- Overfitting is most closely associated with:
 - A. Too simple a model for a complex task
 - B. Model memorizing training data and performing poorly on new data
 - C. Too high the learning rate
 - D. Infinite training time

Question 13

- What role does the bias term play in linear models and neural nets?
 - A. Forces the decision boundary through the origin
 - B. Shifts decision boundaries and adds flexibility
 - C. Prevents overfitting
 - D. Normalizes inputs

Question 14

- Which learning rate matches each graphical representation?

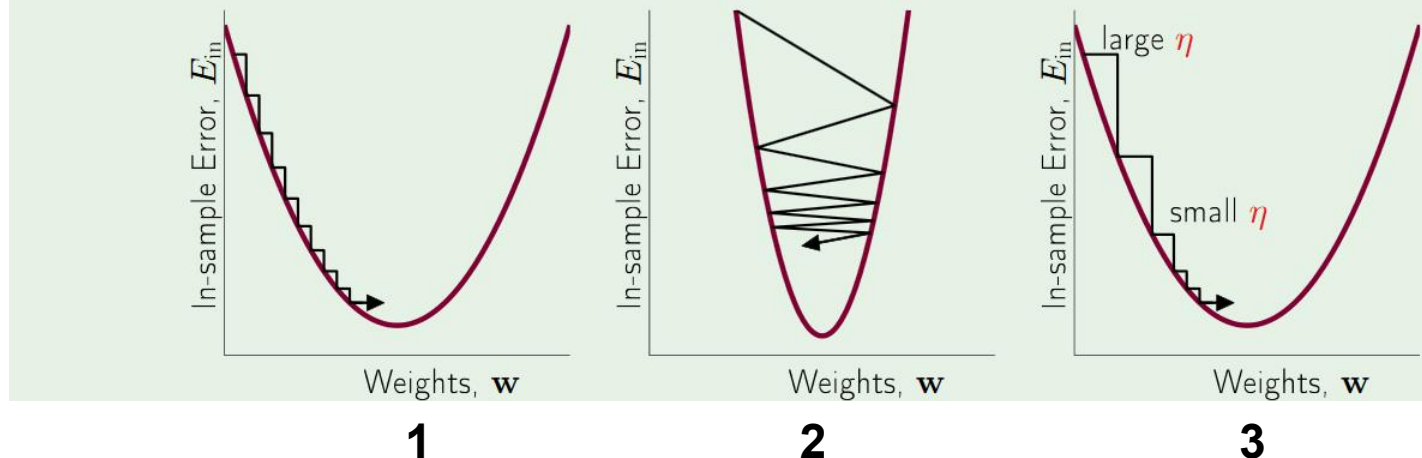
A. η too small

B. η too large

C. η just right

$$w_j := w_j - \eta \frac{\partial E_{in}(\mathbf{w})}{\partial w_j}$$

How η affects the algorithm:



Question 15

- Which validation scheme is the most extreme form of cross-validation?
 - A. 5-fold CV.
 - B. Leave-one-out (LOO).
 - C. Hold-out.
 - D. Bootstrap only.

Question 16

- With very small datasets, which evaluation strategy could be appropriate?
 - A. 5-fold CV.
 - B. Leave-one-out (LOO).
 - C. Hold-out.
 - D. No validation.

Question 17

- When dealing with very large models and datasets, which evaluation strategy is the most commonly used?
 - A. 5-fold CV.
 - B. Leave-one-out (LOO).
 - C. Hold-out.
 - D. No validation.

Question 18

- "Data snooping" refers to:
 - A. Using only training data for all decisions
 - B. Any leakage of test information into modeling decisions
 - C. Hyperparameter tuning on the validation set
 - D. Random initialization of weights

Question 19

- Which of the following situations could be considered data snooping:
 - A. Someone calculates the mean and standard deviation to standardize the data using all the examples (i.e. prior to any data partition).
 - B. Our data, when visualized, we have the impression that it's linearly separable and therefore we decided to use the Perceptron Learning Algorithm (PLA).
 - C. We have 3 candidate hypotheses (SVM, MLP, RL), and we decide which one is the best by looking at its performance in the test set.
 - D. All of the above

Question 20

- Which pair below correctly distinguishes loss function vs evaluation metric?
 - A. Loss: measures generalization; Metric: drives optimization
 - B. Loss: optimized during training; Metric: used to evaluate performance
 - C. Loss: computed only on test; Metric: only on train
 - D. Both are always identical

Question 21

- What is our actual goal in machine learning?
 - A. Minimizing training loss
 - B. Reaching zero gradient
 - C. Generalizing to unseen data
 - D. Maximizing parameter count

Question 22

- Feature/variable scaling primarily aims to:
 - A. Make variables integer-valued
 - B. Put variables on similar ranges to speed optimization
 - C. Reduce the need for cross-validation
 - D. Put variables on similar ranges to better understand their meaning

Question 23

- Which coefficients should we use to scale the test data?
 - A. Those calculated in test
 - B. Those calculated before in training
 - C. It's the same

Question 24

- Assuming the following definitions:
 - Batch: set of examples in which the training set is divided (to train with and adjust the weights).
 - Epoch: each time the network, during training, sees the entire training set.
 - Iteration: number of batches required to complete an epoch (number of weights update per epoch).

If we have a dataset with 10.000 images and we decide to use minibatches of 8 images, how many iterations will take to complete an epoch?

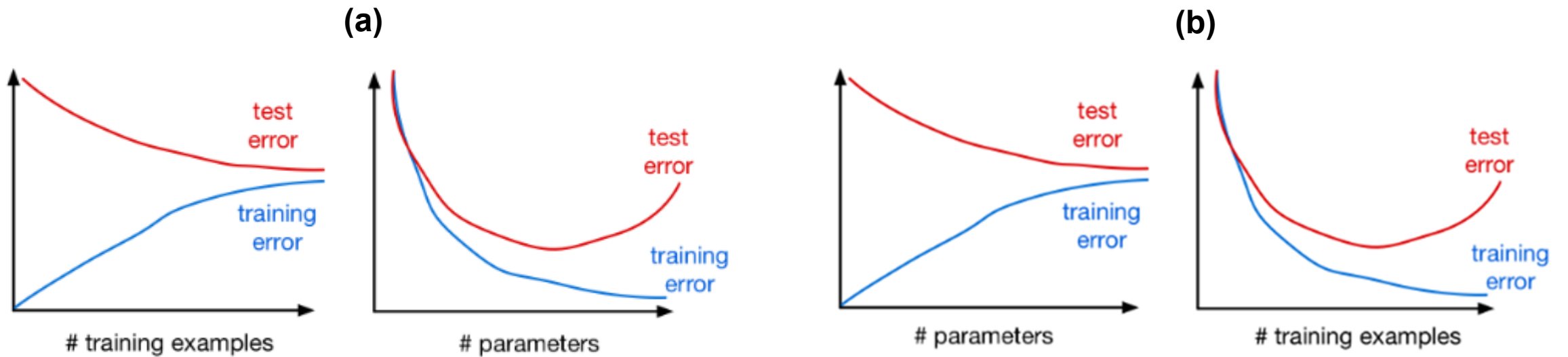
- A. 1
- B. 10
- C. 1000
- D. None of the above

Question 25

- Which statement about model choice vs data type is correct?
 - A. Deep learning is usually the best option for tabular data
 - B. For tabular data, tree-based models often have the right inductive bias
 - C. KNN dominates image tasks
 - D. An autoencoder is best for graphs

Question 26

- Which of the following pairs of graphs is correct?



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